

Menota Lemmatisation Assistant (MLA) 2.0

A user guide

Odd Einar Haugen, 12 November 2023

The first version of MLA was developed in the period 2005–2006 by Paul Meurer at the University Library in Bergen. This second version, also developed by Paul Meurer, is greatly expanded and is no longer just a tool for lemmatisation (and morphological annotation) of Old Norse texts but also an efficient tool for adding a normalised and even a facsimile level to diplomatic texts. Note that there are a few requirements for MLA 2.0 to work:

- A valid XML file according to the Menota schema. An XML editor like Oxygen is recommended, since it offers a continuous checking of the validity of the file.
- MLA is primarily developed for files which contain a diplomatic level of the text and perhaps also a facsimile level, but which lack the normalised level and morphological annotation (for lemma and grammatical forms). The texts from *Gammelnorsk Ordboksverk* in the archive are kept on a single, diplomatic level, while texts from the Menotec project usually contain a facsimile level in addition, but no normalised level.
- Each file must have **xml:id**'s for all words (in the **<w>** element) and punctuation marks (in the **<pc>** element).
- The file must be uploaded to the Menota repository, and the catalogue must subsequently be updated (see ch. 7 below).

1. Workflow – getting in and out of MLA

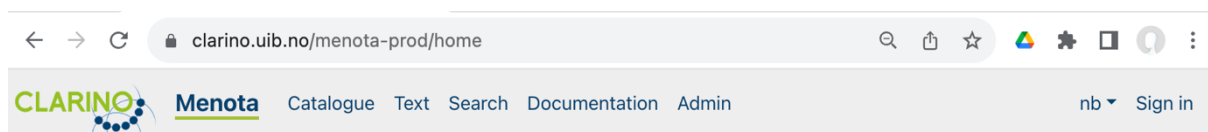
An XML file can be downloaded at any time from MLA and inspected in a suitable XML Editor. This will allow the annotator to check the appearance of the file and to add, delete or revise any part of the encoding. For example, one may want to add **<pc>** elements for the normalised level, since punctuation usually varies a lot between the manuscripts and any normalised edition. This can be done directly in the XML file. When the file is uploaded and updated, any changes will have full effect.

NB! If a **<w>** or **<pc>** element is added in the XML file, this must be given an **xml:id** attribute. The sequence of **xml:id**'s are in the order of 100, e.g. w100, w200, w300, etc. Any addition should be given in steps of 10, e.g. w210 and w220, if two elements are added between w200 and w300. This allows for large additions. If a **<w>** or **<pc>** element is deleted, there is no need to change the **xml:id**'s of the file. The only consideration is that there cannot be two or more identical **xml:id**'s in the file. The **xml:id**'s are assigned on a file by file basis, so there may be a number of w100 **xml:id**'s in the whole corpus.

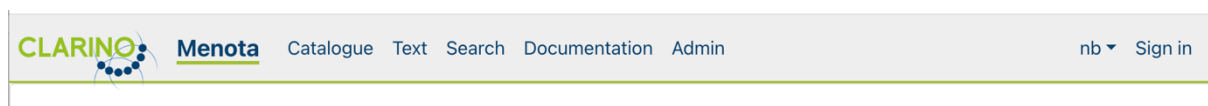
Especially when working with a large file, the annotator will download it from MLA and upload & update it a number of times. As will be seen from the guidelines below, the work may be done in batches, for example first normalising the file, then annotating for word class only, then for the whole range of grammatical categories, and finally, perhaps, adding a facsimile level. Some might like to go “all in” and do the normalisation and morphological annotation in one go.

2. Getting in

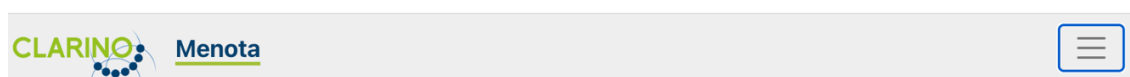
2.1 Use the browser **Chrome** and open <https://clarino.uib.no/menota-prod/home>



2.2 **Sign in** (to the far right in the menu bar) to access MLA:

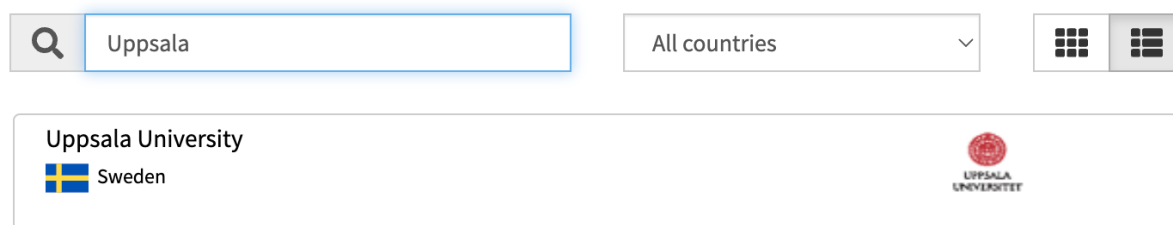


On a narrow screen, **Sign in** will be found in a collapsible menu to the right:

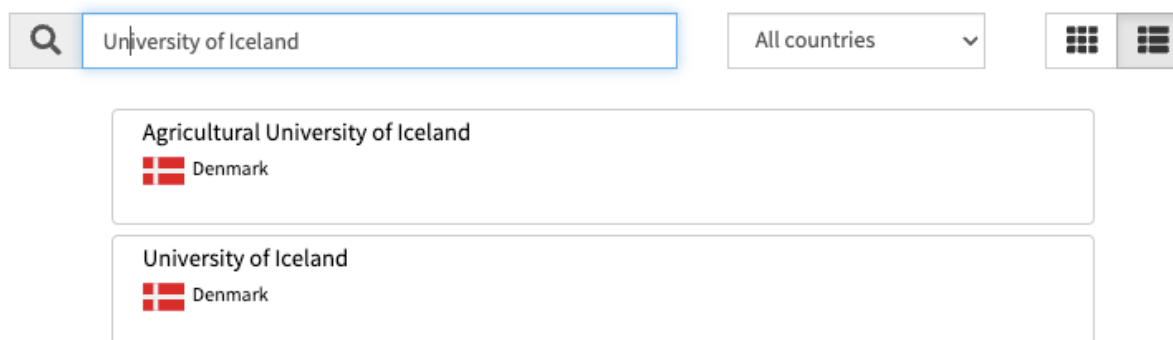


For Norwegian users, **Feide** is probably the simplest option. Under **Tilhørighet** (affiliation) select your affiliated institution, e.g. Universitetet i Bergen. Click on **Fortsett** and select **Bruk arbeids- eller skolekonto**, which will allow you to enter MLA.

Foreign users must find their own institution on the long list offered:



Icelandic users may be surprised to see a Danish flag, but the link should work anyhow:



If no affiliated institution is accessible, it is possible to register via the Clarin website:

All countries

clarín.eu website account

European Union

2.3 You are now in. Click **Catalogue** to access one of the four archives in Menota. Here, the test catalogue has been selected. It is not a huge catalogue, so you can easily leaf through the files.

Corpus:

Test catalogue

This is a test catalogue for Menota texts. Texts in this catalogue should not be quoted or referred to in scholarly publications. Texts are displayed on one or more levels: facsimile (very close transcription), diplomatic (less close and with some interpretation), normalised (regularised orthography). Some texts are morphologically annotated and thus allow for more detailed linguistic searches. A search can be performed on a single text or on any combination of texts.

27 texts.
Filter:
by title/signature
by document
Page:

<

1

2

>

Please note that not all texts in the test catalogue have been indexed (as required by MLA), so for the time being, they cannot be edited in MLA.

2.4 Let us have a look at a short text, a Norwegian diploma from 1266, with a total of no more than 164 words, located in the **Test catalogue**. This text has been edited, normalised and translated in ch. 5 of *Handbok i norrøn filologi* (Bergen: Fagbokforlaget, 2013, pp. 260–261), in case you would like to have a second opinion on your work. Click on the title in the second column to open it:

DN 2:13	1266 : Bergen	Språksamlingane	GNO	0.8.1	1266	nor	164
test version							

2.5 You are now ready to go, whether you would like to normalise or annotate. These procedures are explained in ch. 4, 5 and 6 below. However, before diving into the text, read ch. 3 on how to navigate in MLA.

Navigation is done almost exclusively with keystrokes and arrow keys, and a minimum of mouse clicks. MLA 2.0 puts great weight on ergonomics.

3. Navigating

3.1 After having opened the selected text, this menu bar will be displayed:

DN 2:13 test version – 1266 : Bergen

Page: |< < 1r > >| ☒ Dipl ☐ Norm ☐ Clean text

[Bookmarks](#) [Set bookmark](#) [Reset matches](#) [Parse page](#) ☐ POS only [Download XML](#)

Decide which page you will be working on. In a charter, there are usually only two pages, **1r** for the recto (front) page and **1v** for the verso (back) page.

For longer, multi-page texts (typically a codex), you might want to go to a specific page, depending on your work flow. Then, you must return to **Text** in the menu bar and select the next page you would like to edit.

3.2 Click **POS only** (unless you want to do a full annotation) in order to keep the screen less cluttered. Then click **Parse page** which opens the doors to MLA. This screen now appears:

 **Menota** [Catalogue](#) [Text](#) [Search](#) [MLA](#) [Documentation](#) [Admin](#) nb ▾ [Odd Einar Haugen](#)

DN 2:13 test version – 1r

< 1 2 3 4 5 ... 10 >

Go to page:

[Reparse](#)

[Redisambiguate](#)

Show: ☒ Norm ☐ Facs

[Edit normalization](#)

[Edit facs rules](#)

☐ Show used CG rules

☒ Collapse readings

☒ Key shortcuts w/o Ctrl

☐ Show context box

Dipl	Norm	MSA
<pb ed="ms" n="1r"/>		
<lb ed="DN" n="1"/>		
<lb ed="ms" n="1"/>		
Th{orgils}	[?] Thorgils	1 ?? unknown
m{eð}	[?] með	3 með xAP [6058] ...
gvðs	[?] guðs	1 guð xNC [10]
miskun{n}	[?] miskunn	2 miskunn xNC [136] ...
byskop	[?] byskop	1 biskup xNC [15]
i	[?] i	7 i xAP [8913] ...
stawang{re}	[?] stawangri	1 ?? unknown
.	.	1 . xPC
sender	[?] sendir	1 senda xVB [40]
<lb ed="DN" n="2"/>		
lærð om	[?] lærð um	2 lærðr xAJ [1] ...

Begin by finding your point of departure. Within each page there will be a number of screens that follow each other. For a short text like DN 2:13, there are no more than 10 screens, but

for larger texts, there will be many, perhaps as many as one for each line of the page. For a charter of 164 words, there will be approx. 15–20 words on each screen.

By default, MLA will open with the first screen of the lowest page number.

The meaning of the boxed question mark as well as the proposed lemma in the third column will be explained in ch. 4.1 below.

3.3 Make some initial decisions on how you would prefer to have your screen:

Select **Show used CG rules** if you would like to see the Constraint Grammar rules that are applied. This is probably something you would postpone.

Click **Norm** if you are planning to normalise the text, **Facs** if you are planning to add a facsimile level (probably something you would consider at a later stage, see ch. 5 below).

Click **Collapse readings** if you do not want to see the full list of possible grammatical forms (also something you might want to do at a later stage, see ch. 4 below).

Click **Key shortcuts w/o Ctrl** if you would like as few key combinations as possible (recommended). Many commands can then be done by pressing a single key, rather than the **ctrl** key + a specific key.

Click **Show context box** if you want to have the full text visible in the right half of the screen. Try this option and see what you prefer.

3.4 Start moving around

Move from one screen to the next by **ctrl-shift-arrow-right** and to the previous one by **ctrl-shift-arrow-left**. For these commands, the **ctrl**-key must always be used.

Put the cursor in one of the lines. Use the **arrow** keys to move **upwards** or **downwards** on the screen.

3.5 Start interacting with the text. Select a line, for example the first one:

Th{orgils}	? Thorgils	1 ?? unknown
------------	-------------------------	---------------------------

Click **m** (or **ctrl-m** if you have not ticked the box for **Key shortcuts w/o Ctrl**), and you will be able to edit the normalised form in a box that now appears. More on this in ch. 4 below.

Click **arrow-right** (or **ctrl-arrow-right**) to be able to edit the lemma and grammatical information of the word. More on this in ch. 6 below.

3.6 Change the delimitation of words. MLA will import the **<w>** and **<pc>** elements from the file, displaying them line by line. For normalisation as well as lemmatisation, it is sometimes necessary to join or split **<w>** elements. This can be done from inside the MLA application.

3.6.1 Joining words. The words “salo” and “hjalpar” should be joined into a single word “salohjalpar”, as most dictionaries would prescribe (see e.g. “sáluhjǫlp” in ONP):

salo	☐ salu	3	sála	xNC	[27] ...
hjalpar	☐ hjalpar	2	hjǫlp	xNC	[24] ...

Select the first word, then click **j** (or **ctrl-j**) for “join”, and MLA will do its magic, leaving a small joining symbol in red:

salo hjalpar	→	☐ salo hjalpar	1	sáluhjǫlp	xNC	[2]
--------------	---	---------------------------------------	---	-----------	-----	-----

The two words “salo hjalpar” have now been joined into a single word (although with an internal space), and this can be normalised and annotated as a single word. More on these procedures in ch. 4 and 6 below.

This is what the word looks like in the XML file which MLA will export:

```
<w xml:id="w9700">
  <choice>
    <me:dipl>salo hjalpar</me:dipl>
    <me:fac>&slong;alo &hrdes;&inodot;alpar</me:fac>
    <me:norm>salo hjalpar</me:norm>
  </choice>
</w>
```

The normalised form “salo hjalpar” is not correct; it should be “sáluhjǫlp”. We will return to this in ch. 4 below.

Ouch! You have joined two words which should not have been joined. Select the word, then type **j** (or **ctrl-j**) again, and the two words are re-instated.

3.6.2 Splitting words. The two words “viðr komandum” are encoded with a space between them within a single **<w>** element:

viðr komandom	☐ viðr komandum	1	viðrkomandi	xAJ	[2]
---------------	--	---	-------------	-----	-----

In order to split them into “viðr” and “komandum”, select the line and click **y** (or **ctrl-y**). Think of **y** as a character that is split into two parts. This screen now appears:

Split word



... <pb/> <lb/> <lb/> Th{orgils} m{eð} gvðs miskun{n} byskop i stawang{re} . sender <lb> lærð|om </lb> {ok} lendum verandom {ok} **viðr komandom** ollvm gvðs vínun {ok} <lb/> sínun <lb/> þæim sem þetta bræf sia eða hœyra Queðiu Guðs ok sína . ver vilium ollun <lb/> monnom kunnikt gera at ...

Word: **viðr komandom**

Facs: vıð&rrot; koman&drot;om

Dipl: viðr komandom

Cancel

OK

Add a # between the two words. In this case the # will replace the space, if the word to be split does not have an internal space, just add the # where it should be split:

Split word



... <pb/> <lb/> <lb/> Th{orgils} m{eð} gvðs miskun{n} byskop i stawang{re} . sender <lb> lærð|om </lb> {ok} lendum verandom {ok} **viðr komandom** ollvm gvðs vínun {ok} <lb/> sínun <lb/> þæim sem þetta bræf sia eða hœyra Queðiu Guðs ok sína . ver vilium ollun <lb/> monnom kunnikt gera at ...

Word: **viðr komandom**

Facs: vıð&rrot;#koman&drot;om

Dipl: viðr#komandom

Cancel

OK

After having clicked **OK**, the line has been split in two, and there is a small splitting symbol in red as a reminder. The two words can now be edited separately:

```
<seg>
viðr  ↗  [?] viðr      7  viðr xAP [165] ...
komandom  [?] komandum  1  ?? unknown
</seg>
```

In the XML text exported from MLA, the split words will be encoded according to the rules of Menota (cf. ch. 5.3.2). Note that MLA automatically updates the **xml:id** of the second word:

```

<seg type="nb">
  <w xml:id="w1500">
    <choice>
      <me:dipl>viðr</me:dipl>
      <me:facs>v&inodot;ð&rrot;</me:facs>
      <me:norm>viðr</me:norm>
    </choice>
  </w>
  <w xml:id="w1510">
    <choice>
      <me:facs>koman&drot;om</me:facs>
      <me:dipl>komandom</me:dipl>
      <me:norm></me:norm>
    </choice>
  </w>
</seg>

```

There is no normalised form of the second word. We will deal with this in ch. 4 below.

Ouch! You have split a word which should not have been split. With the first line selected, type **y** (or **ctrl-y**) again, and you are offered the possibility of undoing the splitting:



Click **OK** and you are back on track.

Finally, disregard any elements in a field when editing a word. Here, the **<lb>** element states that there is a linebreak within the word:

```
far<lb n="9" ed="DN"/>sældar
```

Do not change anything here, just insert the **#** where you would like to split the word:

```
far<lb n="9" ed="DN"/>#sældar
```

Ouch (again)! It was not a good idea to split this word (it really is a single word), so you may retrace your steps as explained above.

4. Normalising and lemmatising a text

4.1 After having opened the selected text, in this case **DN 2:13 test version**, and having ticked **POS only**, the screen will open with three columns.

The first column is the text on a diplomatic level, in which expanded abbreviations are displayed in curly brackets.

The second column contains a proposed normalisation.

The third column offers a proposed lemma and its word class (see the *Menota Handbook* ch. 11.6.3 for a recommended list of word classes). The proposed lemmata are based on the whole Menota corpus and are surprisingly often correct, although with a notable exception for proper names.

Th{orgils}	☐ Thorgils	1 ?? unknown
m{eð}	☐ með	3 með xAP [6058] ...
gvðs	☐ guðs	1 guð xNC [10]
miskun{n}	☐ miskunn	2 miskunn xNC [136] ...

A small number at the opening of the third column indicates the number of alternatives.

4.2 Begin by normalising the text, which is done by editing the second column. If the proposed normalised form is correct, type **s** (or **ctrl-s**) in the selected line, and a check mark will replace the question mark, indicating that the normalisation has been accepted. Move from one line to another with the **arrow-down** (or **arrow-up**) key.

Here, the normalisation of the second, third and fourth word have been accepted and saved to the MLA file:

Th{orgils}	☐ Thorgils	1 ?? unknown
m{eð}	✓ með	3 með xAP [6058] ...
gvðs	✓ guðs	1 guð xNC [10]
miskun{n}	✓ miskunn	2 miskunn xNC [136] ...

For the first word, you would probably like to have “Porgils” rather than “Thorgils”. Select this line and type **m** (or **ctrl-m**) to open an edit window:

Add norm/facs



... <pb/> <lb/> <lb/> **Th{orgils}** m{eð} gvðs miskun{n} byskop i stawang{re} . sender <lb> lærð|om </lb>
 {ok} lendum verandom {ok} <seg> viðr komandom </seg> ollvm gvðs vnum {ok} <lb/> sínum <lb/>
 þæim sem þetta bræf sia eða hœyra Quediu Guðs ok sina . ver vilium ollum <lb/> monnom
 kunnikt gera ...

Dipl: Th{orgils}

Dipl XML: Th<ex>orgils</ex>

Norm XML:

Cancel

OK

Type “Þorgils” in the open field. There are no proposed forms here, but in many cases there will be two or more alternatives, which you can select simply by clicking. Remember to press **OK** when leaving the window.

Th{orgils}	✓ Þorgils	1 ?? unknown
m{eð}	✓ með	3 með xAP [6058] ...
gvðs	✓ guðs	1 guð xNC [10]
miskun{n}	✓ miskunn	2 miskunn xNC [136] ...

4.3 Continue (if you like) by lemmatising the text, i.e. adding or correcting the lemma (dictionary entry) as well as the word class (part of speech, **POS**).

The four initial words of the charter have now been normalised, but the lemma and word class are missing (or might be wrong) for the first word. Having selected this line, press **arrow-right** (or **ctrl-arrow-right**) in order to select the word for annotation. Press **m** (or **ctrl-m**) in order to open a new edit window:

Add reading



... <pb/> <db/> <db/> **Th{orgils}** m{eð} gvðs miskun{n} byskop i stawang{re} . sender <db> lærd{om} </lb>
 {ok} lendum verandom {ok} <seg> viðr komandom </seg> ollvm gvðs vínun {ok} <db/> sínun <db/>
 þæim sem þetta bræf sia eða hœyra Queðiu Guðs ok sína . ver vilium ollun <db/> monnom
 kunnikt gera ...

Dipl: **Th{orgils}**

Dipl XML: **Th<ex>orgils</ex>**

Norm XML:

Lemma:

☐ empty lemma and features

Features:

POS: **xVB** verb · **xNC** noun, common · **xNP** noun, proper · **xAJ** adjective · **xAV** adverb, general ·
xAQ adverb, interrogative · **xAP** preposition · **xCC** conjunction · **xCS** subjunction ·
xRP relative pronoun · **xPE** pronoun, personal · **xPR** pronoun, reflexive ·
xPQ pronoun, interrogative · **xPI** pronoun, indefinite · **xDD** determiner, demonstrative ·
xDO determiner, quantifier · **xDP** determiner, possessive · **xNA** numeral, cardinal ·
xNO numeral, ordinal · **xIM** infinitive marker · **xVP** verbal particle · **xIT** interjection ·
xPD pronoun, general · **xNU** numeral, general · **xCU** conjunction, general · **xAT** article ·
xNE enclitic · **xEX** expletive particle · **xFW** foreign word · **xFN** foreign name · **xUA** unassigned

Cancel

OK

The normalised form “Þorgils” has been saved, as stated in the **Norm XML** field, but you have to add the lemma. That would be the nominative case of the word, which is identical with the normalised form, “Þorgils”. However, we suggest that you follow ONP and use an initial small letter, “þorgils”.

The next step is to assign word class, which is **xNP** (proper noun). So this is what the edit window should look like:

Add reading



... <pb/> <db/> <db/> **Th{orgils}** m{eð} gvðs miskun{n} byskop i stawang{re} . sender <db> lærd{om} </lb>
 {ok} lendum verandom {ok} <seg> viðr komandom </seg> ollvm gvðs vínun {ok} <db/> sínun <db/>
 þæim sem þetta bræf sia eða hœyra Queðiu Guðs ok sína . ver vilium ollun <db/> monnom
 kunnikt gera ...

Dipl: **Th{orgils}**

Dipl XML: **Th<ex>orgils</ex>**

Norm XML:

Lemma:

☐ empty lemma and features

Features: xNP

POS: **xNP** noun, proper

Cancel

OK

Roman numerals should not be lemmatised, so for these you tick the box “empty lemma and features”. See the *Menota Handbook* ch. 11.6.3 on Roman numerals.

The first word now appears with a check mark against the normalised form as well as against the lemma and its word class.

Th{orgils}	✓ Þorgils	1 ✓ þorgils xNP
m{eð}	✓ með	3 með xAP [6058] ...
gvðs	✓ guðs	1 guð xNC [10]
miskun{n}	✓ miskunn	2 miskunn xNC [136] ...

Save the proposed lemmata (with word classes) for the three remaining words. If necessary, select the third column by **arrow-right** (or **ctrl-arrow-right**), as seen here. For the second word, there are three possible forms, indicated by the number in front of the proposed lemma. The spelling of the word is the same, but it can be assigned to three different word classes, **xAP**, **xVP** and **xAV**. By far, the first word class is the most frequent one (and this is why it has been listed first), having 6058 instances in the whole Menota corpus, while the two next ones are far less common.

Th{orgils}	✓ Þorgils	1 ✓ þorgils xNP
m{eð}	✓ með	3 með xAP [6058] með xVP [109] með xAV [57]
gvðs	✓ guðs	1 guð xNC [10]
miskun{n}	✓ miskunn	2 miskunn xNC [136] ...

Texts in the Menota archive have been annotated in various project with varying rules. In the *Menota Handbook*, ch. 11.6.3 we suggest stricter guidelines in this field. Based on these recommendations, we go for **xAP** (preposition) as the word class, not **xVP** (verbal particle) or **xAV** (adverb). In order to select the form, type **shift** (or **ctrl-shift**). Having done this for the third and fourth word, too, we are told by MLA that the text has been normalised according to the second column and lemmatised according to the third column.

Th{orgils}	✓ Þorgils	1 ✓ þorgils xNP
m{eð}	✓ með	1/3 ✓ með xAP [6058]
gvðs	✓ guðs	1 ✓ guð xNC [10]
miskun{n}	✓ miskunn	1/2 ✓ miskunn xNC [136]

4.4 The proposed normalisation of the text is based on a number of rules developed in July and August 2023. They cover a number of cases, but there will certainly be rules that you might like to add for recurring words. This can be done on a word-by-word basis pressing the **Edit normalization** button.

For example, the form “ollvm” should be normalised to “ǫllum”, “gvðs” to “guðs”, and “vínvm” to “vínvm”.

Edit normalization rules ×

Add rule

dipl: norm:

Add dipl form as 'word' for whole word, 'prefix-' for prefix, '-suffix' for suffix, '-int-' for internal string, '+nonini-' for non-initial string. Add norm replacement without hyphens or plus sign. See rules below for examples.

Rules

Dipl	Norm	Remove
a	á	
adr	áðr	
aðr	áðr	

Click **Add** and (if you like) check that the rule has been added:

Nu	Nú	
Nv	Nú	
ollafr	Óláfr	
Ollum	Ǫllum	
ollvm	ǫllum	
Ollvm	Ǫllum	
or	ór	

Then click **Apply** to make it work, and continue with the two next words in the same way.

Have a look at the explanation below the **Apply** button. It is possible to normalise parts of a word (initial, medial, final) and to specify exceptions from the rule, e.g. that ‘þ’ should be normalised to ‘ð’, apart from in the beginning of a word.

If you think that a rule is wrong or that it is over-generating, you may click the wastebasket button in order to delete it.

After having added new normalisation rules, MLA will become even better at suggesting normalised forms.

4.5 Punctuation is encoded by the <pc> element. When normalising a text, the punctuation usually has to be adjusted considerably. While some codices have too many punctuation marks compared to a normalised text, charters frequently are in need of additional punctuation. We recommend editing the punctuation in the same workflow as the running text, line by line, since whenever a dot (i.e. a full stop) is encoded on the normalised level, the next word should have a capital letter. Note that we do not make any distinction between the dots in manuscripts, which have a rather wide usage, and the full stops in a normalised text. Both are encoded by the same character, U+002E FULL STOP.

Punctuation characters (as well as any words) will be signalled by a boxed question mark if they have not been normalised. This is an example from *Elíss saga ok Rósamundar* in Upps DG 4–7 fol, where the words have been normalised (cf. the check marks), but not the single punctuation character:

stjorn{n}	✓	stjörn	1	stjörn	xNC	[1]
{oc}	✓	ok	1/10	ok	xCC	[15826] <i>select:38</i>
riki	✓	ríki	4	ríki	xNC	[475] ...
.		? .	1	.	xPC	
valld	✓	vald	1	vald	xNC	[175]
{oc}	✓	ok	1/10	ok	xCC	[15826] <i>select:38</i>
<lb ed="ms" n="18"/>						
skipan	✓	skipun	1	skipun	xNC	[57]

Punctuation characters are edited in the same manner as words (see 4.2 above). Open the edit window by pressing **m** (or **ctrl-m**) and make any necessary changes. In this case, from a dot to a comma:

stjorn{n}	✓	stjörn	1	stjörn	xNC	[1]
{oc}	✓	ok	1/10	ok	xCC	[15826] <i>select:38</i>
riki	✓	ríki	4	ríki	xNC	[475] ...
.	✓	,	1	.	xPC	
valld	✓	vald	1	vald	xNC	[175]
{oc}	✓	ok	1/10	ok	xCC	[15826] <i>select:38</i>
<lb n="18" ed="ms"/>						
skipan	✓	skipun	1	skipun	xNC	[57]

4.5.1 Replacing a punctuation character. In the example above, the dot on the diplomatic level was replaced by a comma on the normalised level, since it is an enumeration – “stjörn ok ríki, vald ok skipun”. In the edit window, type a comma in the **Norm XML** field (or click on it in the list of candidates):

... horskir menn . eina fagra sögu <lb/> drengskaps um raustan <lb> riddera skap </lb> ok <lb> </lb>
 hertoga . er stiorn{n} {oc} ríki . valld {oc} <lb/> skipan . {oc} forsio hafðe yfir lande hi{n}s <lb/>
 helga egidíj í <lb> </lb> <lb> </lb> ...

Dipl: .

Dipl XML: .

Norm XML: ,

☐ empty norm

Candidates: , - . - . - . - ! · tíu · ? · ; · : · . . ,

As for the **lemma** in the third column of the MLA view, we recommend that it is left empty for all punctuation marks. The edit window opens after you have selected the third column by **arrow-right** and then pressed **m** (or **ctrl-m**). Now, tick the box for **empty lemma and features**.

... horskir menn . æina fagra saugu . <lb/> dyrlegs drengskaps . um raustan <lb> riddera scap </lb>
 oc lofsæla atgerð . æins <lb> uirðulegs </lb> hertoga . er stiorn{n} {oc} ríki . valld {oc} <lb/> skipan .
 {oc} forsio hafðe yfir lande hi{n}s <lb/> helga egidíj í utsuðr ætt ríkis fracka konungs . þessi <lb>
 hertogi </lb> lifðe sua lengi . at skegg hans blomgaðez <lb> huitum </lb> lockum . sua ...

Dipl: .

Dipl XML: .

Norm XML: ,

☐ empty norm

Candidates: , - . - . - . - ! · tíu · ? · ; · : · . . ,

Lemma:

☒ empty lemma and features

Afterwards, the zero symbol will be displayed in the third column, indicating that there is no lemma for the punctuation character:

stiorn{n}	✓ stjórn	1	stjórn	xNC	[1]
{oc}	✓ ok	1/10	ok	xCC	[15826] <i>select:38</i>
ríki	✓ ríki	4	ríki	xNC	[475] ...
.	✓ ,	1	✓ Ø		
valld	✓ vald	1	vald	xNC	[175]
{oc}	✓ ok	1/10	ok	xCC	[15826] <i>select:38</i>
<lb n="18" ed="ms"/>					
skipan	✓ skipun	1	skipun	xNC	[57]

4.5.2 Deleting a punctuation character. Still in *Elíss saga*, we would probably delete the punctuation character in the sentence “[Guðshræzla ok gæfu góðleikr] stjórnðu öllu hans lífi. með Guðs forsjá”:

stíorn{n}aðu	✓ stjórnðu	1 ✓ stjórna xVB
aullu	✓ öllu	1 ✓ allr xDQ
hans	✓ hans	1/3 ✓ hann xPE [2546]
livi	✓ lífi	1/3 ✓ lífi xNC [13]
.	? .	1 . xPC
<lb n="23" ed="ms"/>		
með	✓ með	1/3 ✓ með xAP [6083]
guðs	✓ Guðs	1/2 ✓ guð xNP [1]
forsio	✓ forsjá	1/3 ✓ forsjá xNC [16]

Select the line with the punctuation mark, press **m** (or **ctrl-m**) and tick the box **empty norm** to the right of the **Norm XML** field.

... ætt ríkis fracka konungs . þessi <lb> hertogi </lb> lifðe sua lengi . At skegg hans blomgaðez <lb> huitum </lb> lockum . sua var han{n} sæll sína daga . At <lb> guðs|ræzla </lb> . oc gæfu góðlæikr . stíorn{n}aðu aullu hans livi . <lb/> með guðs forsio . Meðan hann var ungr oc a æsku alldre <lb/> þa var hann fullnomi allra goðra siða . oc hirðlegs ...
Dipl: .

Dipl XML: .

Norm XML: ☐ empty norm

Candidates: , - . - - - - - ! . tíu . ? . ; . : . . ,

The punctuation mark will be kept on the diplomatic (and facsimile) level, but on the normalised level, there will be no mark.

Staying in the same line, press **arrow-right** and then **m** (or **ctrl-m**), and tick the box **empty lemma and features** to the right of the **Lemma** field.

Dipl XML: .

Norm XML: ☒ empty norm

Candidates: , - . - - - - - ! . tíu . ? . ; . : . . ,

Lemma: ☐ empty lemma and features

Afterwards, the zero symbol will be displayed in the second column (stating that there should be no punctuation character on the normalised level) and in the third column (stating that there is no lemma for the punctuation character).

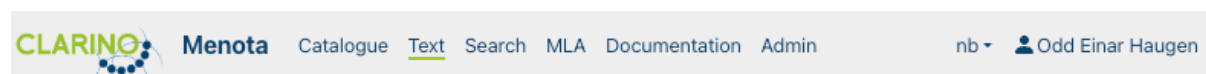
stíorn{n}aðu	✓ stjórnuðu	1 ✓ stjórna xVB
aullu	✓ øllu	1 ✓ allr xDQ
hans	✓ hans	1/3 ✓ hann xPE [2546]
livi	✓ lífi	1/3 ✓ lífi xNC [13]
.	✓ Ø	1 ✓ Ø
<lb n="23" ed="ms"/>		
með	✓ með	1/3 ✓ með xAP [6083]
guðs	✓ Guðs	1/2 ✓ guð xNP [1]
forsio	✓ forsja	1/3 ✓ forsjá xNC [16]

4.5.3 Adding a punctuation character. The addition of punctuation characters will be implemented at later stage, probably by the end of November 2023. For the time being, additions have to be done in the downloaded XML file, which subsequently must be uploaded to the repository and then updated from repository for further work in MLA. Each added punctuation character will be encoded by a new **<pc>** element.

Remember to set a suitable **xml:id** to the new **<pc>** element, as explained on p. 1 above.

4.6 At this point you might want to check the normalisation and annotation in the XML file.

Return to **Text** in the menu bar, and click on **Download XML**:



DN 2:13 test version – 1266 : Bergen

Page: < < 1r > >

☒ POS only

Depending on your browser, you can open the XML file directly from the downloaded files on your computer, and be reassured that the four initial words in the charter DN 2:13 have received the correct normalisation and lemmatisation (so far, the **facts** level is empty):

```
<pb ed="ms" n="1r"/>
<lb ed="ms" n="1"/>
<lb ed="DN" n="1"/>
```

```

<w lemma="þorgils" me:msa="xNP" xml:id="w100">
  <choice>
    <me:dipl>Th<ex>orgils</ex></me:dipl>
    <me:facs></me:facs>
    <me:norm>þorgils</me:norm>
  </choice>
</w>
<w lemma="með" me:msa="xAP" xml:id="w200">
  <choice>
    <me:dipl>m<ex>eð</ex></me:dipl>
    <me:facs></me:facs>
    <me:norm>með</me:norm>
  </choice>
</w>
<w lemma="guð" me:msa="xNC" xml:id="w300">
  <choice>
    <me:dipl>gvðs</me:dipl>
    <me:facs></me:facs>
    <me:norm>guðs</me:norm>
  </choice>
</w>
<w lemma="miskunn" me:msa="xNC" xml:id="w400">
  <choice>
    <me:dipl>miskun<ex>n</ex></me:dipl>
    <me:facs></me:facs>
    <me:norm>miskunn</me:norm>
  </choice>
</w>

```

Remember that in order to have the normalised level displayed, the header must specify that the charter has been encoded on a diplomatic as well as a normalised level:

```

<normalization me:level="dipl norm">
  <p>This text has been encoded on a diplomatic and normalised level.</p>
</normalization>

```

The same applies for any addition of a facsimile level. The **<normalization>** element must be edited manually in the XML file.

4.7 One column at the time? First of all, it should be mentioned that MLA can be used solely for normalisation, in which case you would edit the first column only.

Even if the plan is to normalise and lemmatise, it can be quicker to do one operation at a time. If the proposed normalised forms are to a high degree correct you may prefer to move down line by line and only build a normalised text. As the next operation, the lemmatisation may be done, also going down line by line.

This is a matter of personal preference. Some would like to do the full morphological annotation, too. More on this in ch. 6 below.

5. Adding a facsimile level

5.1 If you are planning on making a multi-level encoding of a file, it is highly recommendable to begin with the facsimile level and then add the diplomatic and possibly the normalised level afterwards. This has been the procedure for the law texts in the Menotec project, which have been encoded on a facsimile and diplomatic level.

However, if you would like to backtrack, as it were, it is possible to add a facsimile level to a file which only has the diplomatic level. In texts with a low degree of abbreviation, such as in most Norwegian (and Swedish and Danish) manuscripts, there is a limited number of conversions to be made. In many Icelandic texts, on the other hand, there is a huge amount of abbreviations. For these texts, adding a facsimile level is daunting.

5.2 Begin by ticking the box for **Facs** after having pressed **Parse page**. You may or may not tick the **Norm** box.

DN 2:13 test version – 1r

< 1 2 3 4 5 ... 10 > Go to page: Reparse Redisambiguate | Show: ☒ Norm

☒ Facs Edit normalization Edit facs rules ☒ Show used CG rules ☒ Collapse readings

☒ Key shortcuts w/o Ctrl ☐ Show context box

The page which now opens offer proposed facsimile readings based on a list of facs rules. They can be edited in the same manner as the normalisation rules:

Rules

Dipl	Facs	Remove
-d-	&drot;	
-dr-	&drot;&rrot;	
-ðr-	ð&rrot;	
-<ex>an</ex>-	<am>&bar;</am>	
-<ex>ann</ex>-	<am>&bar;</am>	
-<ex>er</ex>-	<am>&er;</am>	
-<ex>ir</ex>-	<am>&er;</am>	

There are basically two types of facs rules. One concerns **character variants**, e.g. that an “s” in a diplomatic rendering should be replaced by the tall “l” in a facsimile rendering (if indeed this is the case in the manuscript). Another concerns **abbreviations**, e.g. that “ok” (or “oc”) in a diplomatic rendering should be replaced by the Tironian note “7”.

It will soon be clear that the facs rules are much more dependent on the particular manuscript than the normalisation rules. One should probably begin by analysing the orthography of the manuscript and set up the likely facs rules before beginning the work in MLA. It is essential to have a facsimile of the manuscript in high resolution so as to check the exact orthography and system of abbreviations.

The first words of DN 2:13 tells us that much work remains to be done on the facsimilation of the text:

Dipl	Norm	Facs	MSA
<pb ed="ms" n="1r"/>			
<lb ed="DN" n="1"/>			
<lb ed="ms" n="1"/>			
Th{orgils}	✓ Þorgils	[?] Th<ozgilf	1 ✓ þorgils xNP [1]
m{eð}	✓ með	[?] m<eð>	1/3 ✓ með xAP [6058] <i>select:31</i>
gvðs	✓ guðs	[?] gvðf	1 ✓ guð xNC [10]
miskun{n}	✓ miskunn	[?] mifkun̄o	1/2 ✓ miskunn xNC [136] <i>select:31</i>
byskop	✓ byskup	[?] býfkop	1 ✓ biskup xNC [15]
i	✓ í	[?] i	1/7 ✓ í xAP [8913]
stawang{re}	✓ Stafangri	[?] ftawang<re>	1 ✓ stafangr xNP

5.3 At the moment, the facsimilation of a diplomatic text is in a test phase. Hopefully, further work on the facs rules will lead to a much higher success rate.

6. Full morphological annotation

6.1 So far, we have restricted ourselves to POS only, so that the morphological annotation only includes the lemma (citation form in a dictionary) and its word class (part-of-speech)

The next and final step is to include all grammatical forms for inflected words. For non-inflected words such as prepositions, conjunctions and subjunctions, there is nothing more to add. For inflected word classes, there is a number of categories to fill in, especially for verbs, and with respect to the high degree of syncretism, also for the weak declension of adjectives.

The annotation procedure is the same as described in ch. 4 above, but more time must be spent on the selection of categories and features for a full morphological annotation. The grammatical categories and their abbreviations are explained in the *Menota Handbook* ch. 11.5 and 11.6, and also in the appendix at the end of this document.

Now, the button **POS only** must not be pressed (nor the **Facs** box discussed in the previous chapter). At this point you might on the other hand tick the box **Collapse readings** in order to have a less cluttered screen.

Let us go to the next page in the charter DN 2:13 and have a look at the phrase “þeim sem þetta bréf sjá eða heyra”. This is what the screen looks like with collapsed readings, i.e. just one morphological analysis for each line:

þæím	✓ þeim	3	sá xDD nP gU cD [24] ...
sem	✓ sem	11	sem xCS [1703] ...
þetta	✓ þetta	8	sjá xDD nS gN cA [574] ...
bræf	✓ bréf	4	bréf xNC nP gN cA sI [24] ...
sia	✓ sjá	22	sjá xVB tPS fI vA [100] ...
eða	✓ eða	1	eða xCC [2039]
heyra	✓ heyra	5	heyra xVB tPS fI vA [15] ...

The checkmarks in the second column tell us that this part of the text has been normalised (no more boxed question marks), and the lack of check marks in the third column means that we need to select the correct analysis here.

Well, what do you know! The proposed lemmata in blue are all correct. And the grammatical analysis, too. How come? MLA draws upon the whole annotated Menota corpus, which (if I have counted correctly) amounts to approx. 850.000 words. So the great majority of diplomatic forms have already been annotated somewhere.

However, note that apart from the conjunction “eða” there are more than one alternative for each word. For “þeim” there are three alternatives. After having selected this line, press **arrow-right**, and the whole set opens:

þæím	✓ þeim	3	sá xDD nP gU cD [24]
			sá xDD nS gM cD [5]
			sá xDD nP gM cD [1]

Since this determiner (**xDD**) obviously is in the plural we can discard the second analysis (which, as we know, can be correct in other contexts), while the choice between the first and third analysis is about deciding the gender – either unspecified (**gU**) in the first analysis or masculine (**gM**) in the second. We recommend that for determiners and adjectives in genitive or dative plural, there is no need to specify gender, so go for **gU**. In other words, the first selected analysis is the best.

(I know this from experience. When Jon Erik Hagen and I annotated *Barlaams ok Jósafats saga* in the 1980s, we spent a lot of time trying to decide the gender of these words, contemplating the surrounding context. In the end, we decided that it was a waste of time to

try and decide the gender. There are simply no determiners or adjectives that have discrete gender forms in genitive or dative plural.)

6.2 Let us have a look at the next word, “sem”. How difficult can that be?

sem ✓ sem 11

sem	xCS	[1703]
sem	xRP	[984]
sem	xCS yIN	[611]
sem	xCU yIN	[464]
sem	xCU	[445]
sem	xCC	[245]
sem	xCS ySU	[112]
sem	xCU ySU	[55]
vera	xVB p1 tPS nP fF mSU vA	[3]
vera	xVB p1 tPS nP fF mSU vA iST	[1]

From this list, we understand that “sem” has been classified in a number of ways in the Menota database, which (we must not forget) has been built by a number of people over a number of years by analysing a number of texts. Consistency is hard to achieve.

As recommended in ch. 11.6 in the *Menota handbook*, we should analyse “sem” as a subjunction (**xCS**) in cases like this, disregarding the analysis as relative particle (**xRP**) or conjunction (**xCU**) with or without case governing (**YIN**, **YSU**). However, “sem” might have been a subjunctive form of the verb “vera”, so it is a useful reminder to have this stated. Actually, there are only two alternatives for us, **sem xCS** or **vera xVB p1 tPS nP fF mSU vA** (we should drop the final **iST**; cf. ch. 11.6 of the *Menota Handbook*).

Here, the conclusion is simple: the first analysis is the best one, and as can be seen from the number at the end, also the most frequently chosen one.

6.3 Fine! Let us hunt for a mistake by MLA. This we find two words below.

bræf ✓ bréf 4

bréf	xNC nP gN cA sI	[24]
bréf	xNC nS gN cA sI	[17]
bréf	xNC nP gN cN sI	[14]
bréf	xNC nS gN cN sI	[1]

It is indeed true that the lemma is “bréf”, and that the case (based on the context) is accusative, **cA** (since it is part of a direct object), but it is in the singular, **nS**, not in the plural, **nP**. So we should go for the second analysis offered. Move down to this line by the arrow key, then press **ctrl-return**, and it has been selected:

þetta	✓ þetta	8 sjá xDD nS gN cA [574] ...
bræf	✓ bréf	1/4 ✓ bréf xNC nS gN cA sI [17]
sia	✓ sjá	22 sjá xVB tPS fI vA [100] ...

Have a look at the word above and note that the adjective and the noun have received a similar analysis, **nS gN cA**. A good sign.

6.4 What about words for which MLA have no proposed lemma and grammatical analysis, like “hiauero” on the third page?

hiauero	□ hjaeru	1 ?? unknown
---------	----------	--------------

We might begin by normalising the word, which must be an oblique form, “hjáveru”, of the noun “hjávera”:

hiauero	✓ hjáveru	1 ?? unknown
---------	-----------	--------------

If in doubt, look up the word in ONP – and be pleasantly surprised to see that it has been excerpted by the dictionary:

hjá·vera *sb. f.*

✓ excerpted

✓ citation slips

🕒 citation text

🕒 supplemented

🕒 structured

🕒 definitions in Danish

🕒 definitions in English

Article

Comp., Gloss., Litt., &c.

Citations by ms. date

Citations by source

Search/filter 19 citations...

(see citation slip for text)

með raðe ok hiauero vars virðulegs herra

•NoDipl (1266) 40²⁰

•NoDipl (1266) 40²⁰

From ONP you will note that this seems to be a typical word for Norwegian charters (DN).

Selecting the third column by **arrow-right**, we have to do a bit of manual annotation. Press **m** (or **ctrl-m**) in order to open the **Add reading** window. Begin by typing the lemma “hjávera” in the lemma window (initially filled by question marks).

Dipl XML: **hiauero**

Norm XML:

Lemma:

☐ empty lemma and features

Features:

POS: **xVB** verb · **xNC** noun, common · **xNP** noun, proper · **xAJ** adjective · **xAV** adverb, general · **xAQ** adverb, interrogative · **xAP** preposition · **xCC** conjunction · **xCS** subjunction · **xRP** relative pronoun · **xPE** pronoun, personal · **xPR** pronoun, reflexive · **xPQ** pronoun, interrogative · **xPI** pronoun, indefinite · **xDD** determiner, demonstrative · **xDQ** determiner, quantifier · **xDP** determiner, possessive · **xNA** numeral, cardinal · **xNO** numeral, ordinal · **xIM** infinitive marker · **xVP** verbal particle · **xIT** interjection · **xPD** pronoun, general · **xNU** numeral, general · **xCU** conjunction, general · **xAT** article · **xNE** enclitic · **xEX** expletive particle · **xFW** foreign word · **xFN** foreign name · **xUA** unassigned

Now, move down to features and pick the correct grammatical form. Having selected xNC for common noun, all features relevant for this word class are displayed:

Features: xNC

POS: **xNC** noun, common

Number: **nS** singular · **nP** plural · **nD** dual · **nSP** singular or plural · **nU** unspecified number

Gender: **gM** masculine · **gF** feminine · **gN** neuter · **gMF** masculine or feminine · **gMN** masculine or neuter · **gFN** feminine or neuter · **gMFN** masculine or feminine or neuter · **gU** unspecified gender

Case: **cN** nominative · **cA** accusative · **cD** dative · **cG** genitive · **cAD** accusative or dative · **cAG** accusative or genitive · **cGD** genitive or dative · **cAN** accusative or nominative · **cO** oblique (i.e. non-nominative) · **cB** ablative · **cV** vocative · **cU** unspecified case

Definiteness: **sl** indefinite · **sD** definite · **sU** unspecified definiteness

Move downwards and click on the correct features. This is a fairly quick process (unless you are not sure about the analysis). This should be the result:

Features: xNC nS gF cD sI

POS: xNC noun, common

Number: nS singular

Gender: gF feminine

Case: cD dative

Definiteness: sI indefinite

Click **OK** when you are satisfied with the lemma and its grammatical form (you may change anything within this window), and see that the word now has received a full morphological annotation:

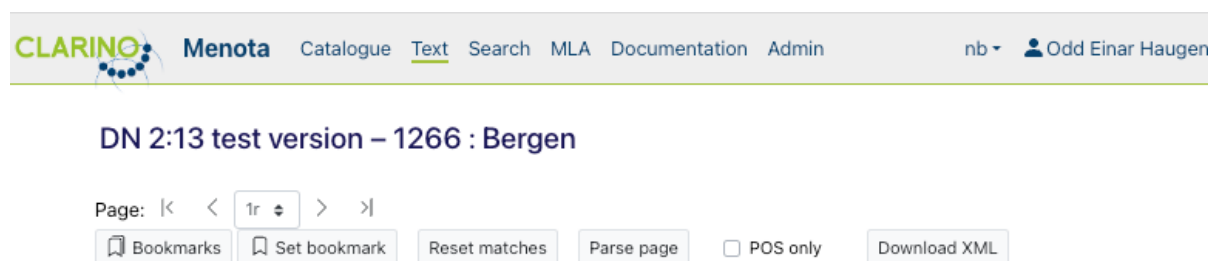
hiauero ✓ hjáveru 1 ✓ hjávera xNC nS gF cD sI

From these examples, it is hopefully clear what to do in other cases.

A particular challenge are proper nouns, whether of once living persons or of fictive ones, place names, existing or fictive. Once a decision has been made, the actual encoding in MLA should be straight forward, though.

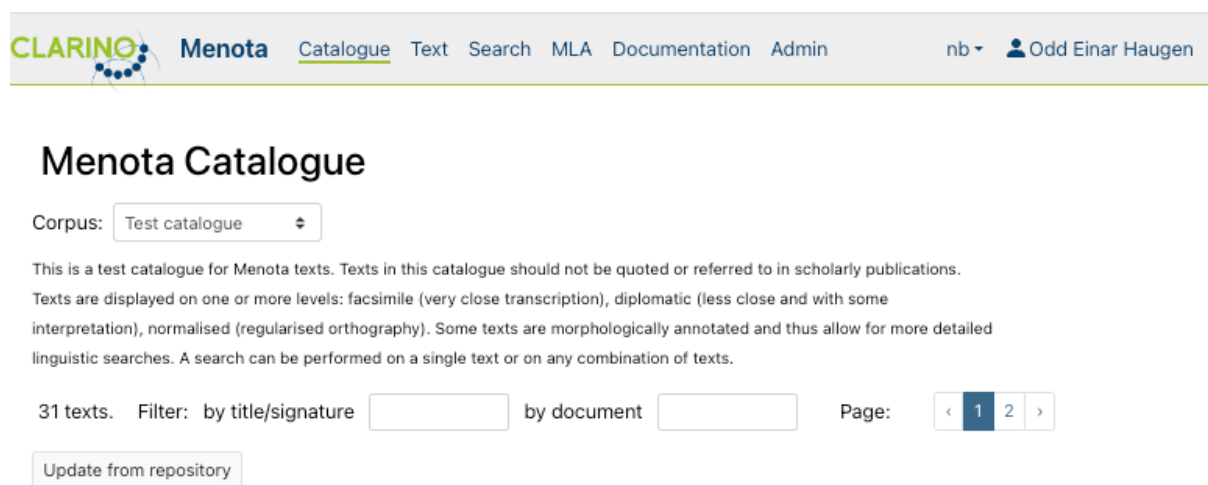
7. A final word on the workflow

As stated in ch. 1 above, when working with a text, you should expect to switch between MLA and the downloaded version in XML. When downloading a text, go to **Text** in the menu bar and press **Download XML**:



After having downloaded a text in XML and edited it in any way you like, it must be uploaded to the Menota repository if you would like to see it on screen or continue working on it. For the time being, you may contact **odd.haugen@uib.no** for the uploading of texts in the Menota repository, while we are planning how to administer the necessary rights.

Assuming you have the necessary rights, go to **Catalogue** in the menu bar and press **Update from repository**. It may take a few minutes for the update to be effected.



Remember that if a normalised level has been added, this must be stated in the **<normalization>** element of the header:

```
<normalization me:level="dipl norm">
  <p>This text has been encoded on a diplomatic and normalised level.</p>
</normalization>
```

Otherwise, the normalised level will not be displayed in the online Menota archive.

Appendix: Abbreviations for word classes and morphological categories

A. Word Classes

xNC	noun, common
xNP	noun, proper
xAJ	adjective
xPE	pronoun, personal
xPR	pronoun, reflexive
xPQ	pronoun, interrogative
xPI	pronoun, indefinite
xDD	determiner, demonstrative
xDQ	determiner, quantifier
xDP	determiner, possessive
xVB	verb
xAV	adverb, general
xAQ	adverb, interrogative
xAP	preposition
xCC	conjunction
xCS	subjunction
xIM	infinitive marker
xIT	interjection
xUA	unassigned
xFW	foreign word

B. Morphological Categories

p1	person: 1. person
p2	person: 2. person
p3	person: 3. person
pU	person: unspecified
nS	number: singular
nP	number: plural
nU	number: unspecified
tPS	tense: present
tPT	tense: preterite
tU	tense: unspecified
mIN	mood: indicative
mSU	mood: subjunctive
mIP	mood: imperative
mINSU	mood: indicative or subjunctive
mINIM	mood: indicative or imperative
mSUIM	mood: subjunctive or imperative

mU	mood: unspecified
fF	finiteness: finite
fI	finiteness: infinitive
fP	finiteness: participle
fU	finiteness: unspecified
vA	voice: active
vR	voice: reflexive
vU	voice: unspecified
gM	gender: masculine
gF	gender: feminine
gN	gender: neuter
gMF	gender: masculine or feminine
gMN	gender: masculine or neuter
gFN	gender: feminine or neuter
gMFN	gender: masculine, feminine or neuter
gU	gender: unspecified
cN	case: nominative
cG	case: genitive
cD	case: dative
cA	case: accusative
cAD	case: accusative or dative
cGD	case: genitive or dative
cAN	case: accusative or nominative
cO	case: oblique
cU	case: unspecified
rP	degree: positive
rC	degree: comparative
rS	degree: superlative
rU	degree: unspecified
sl	definiteness: indefinite
sD	definiteness: definite
sU	definiteness: unspecified